

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 10 · LECTURE NOTES

Voting and Social Choice

The impossibility results, and what they actually forbid.

MODULE OUTCOMES

Describe the standard voting rules and their behaviour.

State Arrow's theorem precisely.

State Gibbard-Satterthwaite and what follows.

Explain the computational escape and its limits.

Evaluate a voting rule honestly.

1 The rules

Rule	How it works	Its characteristic problem
Plurality	The option with the most first-place votes wins.	Vote splitting between similar options, and it discards all information below first place.
Borda count	Points assigned by rank position, summed.	Highly manipulable, and adding clones of an option distorts the result.
Condorcet methods	Elect the option that beats every other in a pairwise comparison.	Such a winner may not exist — cycles, see the callout.
Instant runoff / single transferable vote	Eliminate the last-placed option and redistribute, repeat.	Non-monotone: gaining support can cause a candidate to lose. See the note.
Approval voting	Approve any subset; the most approvals wins.	Where a voter sets the approval threshold is itself a strategic decision.

The same set of ballots can elect different winners under different rules — which is easy to demonstrate with five voters — **and is the fact that makes rule choice a substantive political decision rather than a formality.** Instant runoff's non-monotonicity surprises people and is real: there are constructible profiles in which a candidate loses because some voters ranked them higher, which follows from the elimination order changing.

AND A CONDORCET CYCLE SHOWS THE DIFFICULTY IS IN THE PREFERENCES, NOT THE RULE

Three voters and three options: $A > B > C$, $B > C > A$, $C > A > B$ — A beats B two to one, B beats C two to one, and C beats A two to one, in pairwise majority votes.

So the collective preference is cyclic even though every individual's preference is perfectly transitive — which no voting rule can fix, because there is nothing coherent for it to report: the majority relation itself is not an ordering.

Which means the impossibility results of §§2 and 3 are not about clumsy or badly designed rules — they are about what aggregation can do in principle — and that is why they are theorems rather than criticisms.

And it takes only three voters and three options to construct — so this is not a corner case, it is the generic situation whenever preferences are genuinely diverse, which is when voting is needed at all.

2 Arrow

No rule aggregating individual rankings over 3 or more options satisfies unanimity, independence of irrelevant alternatives, and non-dictatorship together.

- **The theorem is about ranking rules** — those producing a full social ordering from individual orderings — which is worth noting, since rules that produce only a winner are covered by §3 instead.
- **Unanimity: if every individual prefers A to B, the social ranking does too** — which nobody wants to give up.
- **Independence of irrelevant alternatives: the social ranking of A and B depends only on the individuals' rankings of A against B**, and not on where C sits — which is the condition that sounds technical and is load-bearing.
- **And non-dictatorship: no single individual's ranking determines the social one regardless of everybody else's** — which also nobody wants to give up.
- **So one of the three must go, and independence is the one usually given up — and giving it up is precisely what admits manipulation**, because a rule whose verdict on A versus B depends on C can be moved by misreporting about C. **Which condition a rule abandons is the useful way to classify it** (§4's first question).

3 Gibbard-Satterthwaite

ANY NON-DICTATORIAL VOTING RULE OVER THREE OR MORE OUTCOMES IS MANIPULABLE

If the rule can elect any of at least three outcomes and is not a dictatorship, then there exists some preference profile in which some voter benefits from misreporting — which is a theorem, not an observation about existing rules, and it follows from Arrow.

So there is no strategy-proof voting rule — and 'this voting rule is strategy-proof' is therefore either false or the rule is restricted in one of the stated ways (fewer than three reachable outcomes, a restricted preference domain, or a dictator).

Which is exactly Module 06 §4's limit in the no-money setting: payments are what make VCG possible, and voting has none, so the ordinal setting is substantially more constrained than the quasilinear one — which is the structural reason money matters so much in mechanism design.

And the escape routes are narrow: restrict the preference domain (single-peaked preferences admit the median voter rule, which is strategy-proof), introduce money, or accept manipulability and manage it — which is what every real system does, mostly without saying so.

4 The computational escape

- The idea: if finding a profitable misreport is NP-hard, then manipulation is impractical even though it is possible — which was this field's most attractive proposal and is why computational social choice exists as a subfield.
- And manipulation is indeed NP-hard for some rules, including some that are actually used, particularly with weighted voters or many candidates — which is a genuine result and was reasonably exciting.
- But NP-hardness is a worst-case notion (CSCE 637 Module 03's framing) — and typical instances are frequently easy, with simple heuristics succeeding on the great majority of profiles drawn from plausible distributions.
- Which is the standard critique and it holds: average-case hardness is what would actually be needed, and it is very much harder to establish — the same gap that limits worst-case hardness as a basis for cryptography (CSCE 711 Module 02).
- So the computational escape is partial — a real obstacle and not a guarantee — which is the honest summary, and is more useful than either the original enthusiasm or a flat dismissal.

SO HOW TO EVALUATE A VOTING RULE

Which Arrow condition does it give up? — almost always independence — and that tells you where it is manipulable, since the manipulation will involve misreporting about an option that is not in contention.

What is its characteristic manipulation? — every rule has one, and it should be stated in the documentation rather than discovered by participants, which is the difference between a known limitation and a scandal.

Does it satisfy monotonicity, and does it elect a Condorcet winner when one exists? — two properties people assume and that several widely used rules lack (§1's table).

And what does it do when there is a cycle? (§1) — because something has to be done, and the answer is a design decision rather than an implementation detail: different Condorcet completion methods resolve cycles differently and defensibly.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Brandt et al. — Handbook of Computational Social Choice (free PDF)	The whole module , free — and the manipulation-complexity chapters cover §4 including the critique.
Arrow — Social Choice and Individual Values	§2 — the original, and the conditions are stated more carefully there than in most summaries. Free in full from the Cowles Foundation, which published it.
Satterthwaite — Strategy-proofness and Arrow's conditions	§3 — and the connection to Arrow is made explicit.
Shoham & Leyton-Brown, chapter 9 (free PDF)	§§1 to 3 , free — the rules and the impossibilities, with worked examples.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Construct ballots** where five rules elect five different winners.
- 2 **Construct a Condorcet cycle** and verify all three pairwise results.
- 3 **Construct a non-monotone instance** for instant runoff.
- 4 **State Arrow's three conditions** in your own words.
- 5 **Show that plurality violates independence**, with an example.

- 6 **Find the manipulation** for Borda, concretely.
- 7 **State Gibbard-Satterthwaite** and its three escape routes.
- 8 **Verify the median rule is strategy-proof** for single-peaked preferences.
- 9 **Implement a manipulation heuristic** for one rule and measure its success rate.
- 10 **Evaluate one real voting rule** against §4's four questions.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name five voting rules and each one's characteristic problem.
- 2 Construct a Condorcet cycle and say what it shows.
- 3 Why are the impossibilities not about clumsy rules?
- 4 State Arrow's three conditions.
- 5 Which is usually given up, and what does that admit?
- 6 State Gibbard-Satterthwaite.
- 7 How does it relate to Module 06's limit?
- 8 Name the three escape routes.
- 9 State the computational escape and the standard critique.
- 10 Give the four questions for evaluating a rule.